

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Programming Fundamentals	Course Code:	CS 1002
Program:	BS (CY)	Semester:	Fall 2025
Due Date	--	Total Marks:	-
Section:	1A	Page(s):	6
Type:	Project	Weightage	10

Important Instructions:

1. Submit your solution named as your roll number, i.e., 25L_1111.cpp. Do not zip your file.
2. You are not allowed to copy solutions from other students. We will check your code for plagiarism using plagiarism checkers. If any sort of cheating is found, negative marks will be given to all students involved.
3. If we find that you have copied code from the Internet, you will get negative marks.
4. Late submission of your solution is not allowed

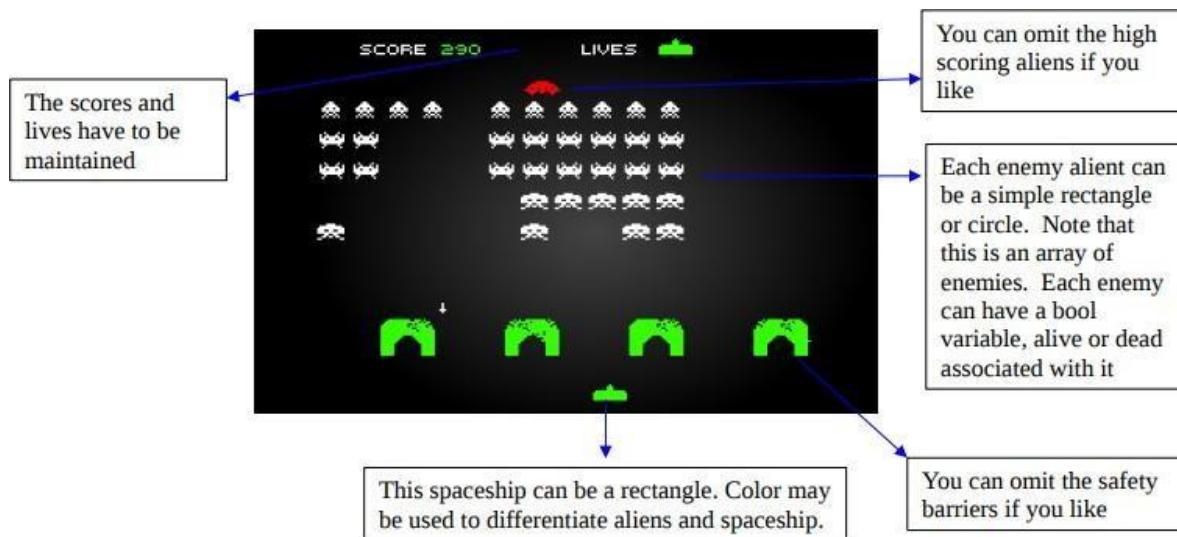
Requirements of PF Project:

1. Colors should be used.
2. You can do your project in groups (A group should not include more than three students).
3. Graphics library is not compulsory to use. However, you can use it if you want.
4. You cannot use any concept not taught in the class, such as dynamic memory, maps, string class, etc.
5. You can use struct.

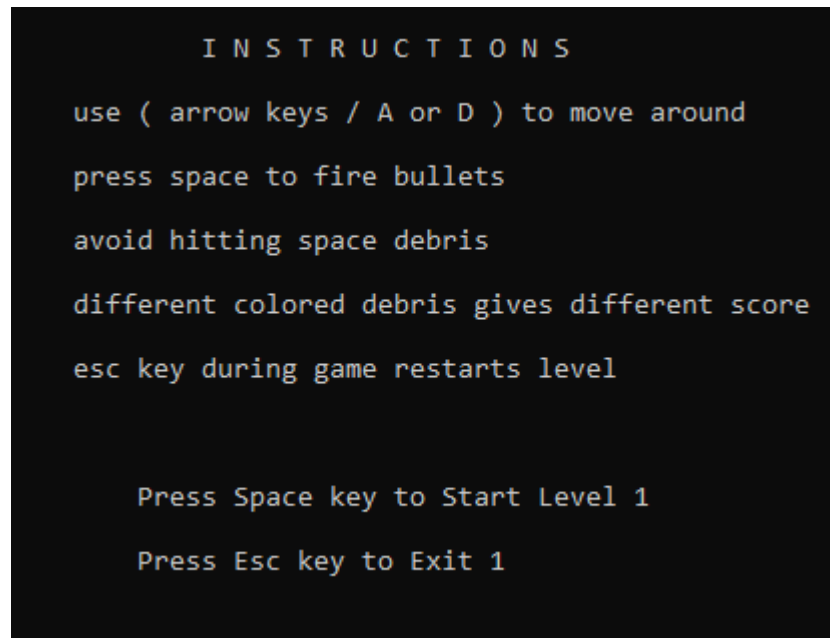
You can make either of these two games.

Option 1:

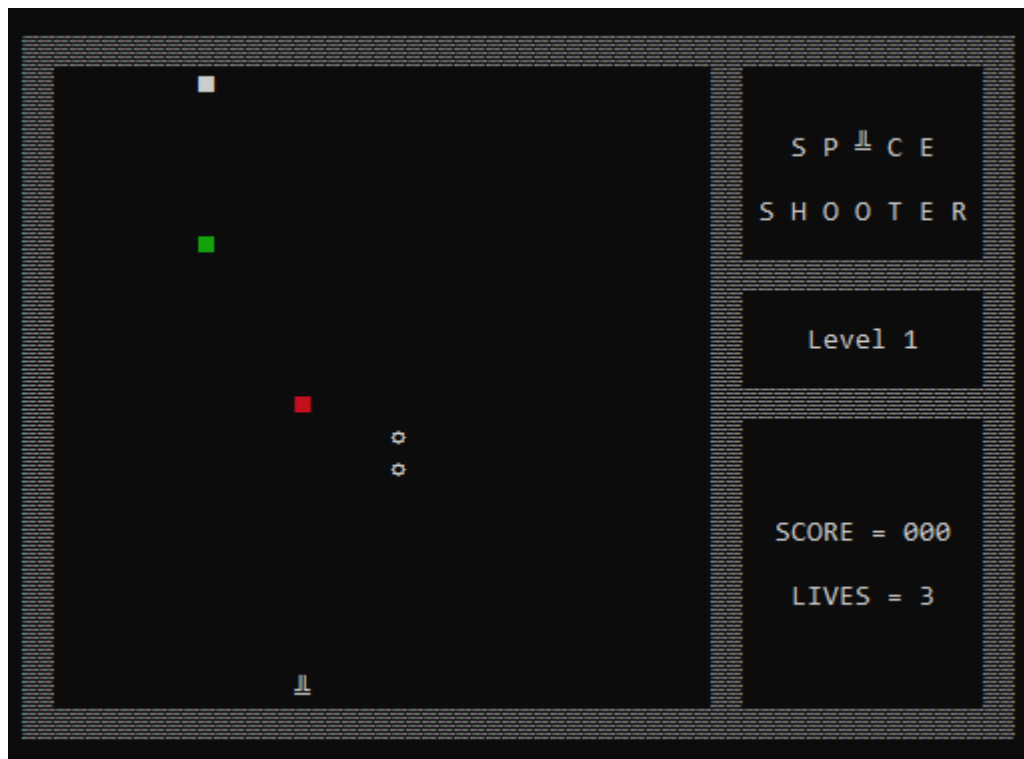
Space shooter is a popular shooting game. In which our galaxy is attacked by alien shooters. The aliens intend to destroy everything on our planet. Your goal is to shoot them all before it is too late.



Your task is to make a simplified version of this game. For Example:



At the start of the game display some playing instructions. After pressing any key like space your game will start like this:



Game Description:

The game includes a spaceship that will fire the bombs when the user hits the space bar. Now it is your choice you can change the buttons. However, if the enemy spaceship hits the user spaceship, he/she will lose one life. There should be three lives in total and when the user loses all of his lives then he will lose the game and he has to start from level 1 again. If the user's score surpasses (level no* 10), the user will proceed to the next level and in the next level there should be more enemy spaceships and the speed of the game should be faster. For each user, you should store his\her highest score in the file. When a user will complete all 5 levels, the user will win the game. After winning the game he/she should have an option to restart from level 1

Option 2:

You will implement a single player game **"Candy Crush"** in C++. Here, a 2-D grid will be used for candies storage and the minimum size of grid will be 9×9.

At start of the game the whole grid will be filled with the new candies by randomly selecting them. The user must be provided with swapping options, so that he can swap two candies, if they lie side by side in same column, or row of grid. The target goal points and number of moves (swaps) should be set at start.

Type of Candies and their corresponding points are listed below.



Plain Candies	Points gained when crushed
 Red - Jelly Bean	30 Points
 Yellow - Lemon Drop	30 Points
 Green - Gum Square	40 Points
 Blue - Lollipop Head	50 Points
 Orange - Lozenge	60 Points
	Color Bomb

Special Candies	
Striped Candies	
    	
Wrapped Candies	
    	

Game Description:

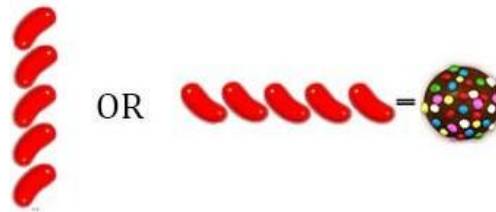
- Crush Plain Candies:** When three similar candies lie in same column, or row of grid, they will be get crushed and empty slots of grid will be refilled with new candies by shifting candies from top towards bottom.
- Striped Candy:** It is created when 4 similar plain candies lie in same column, or row of grid. Striped candy can be made with 5 different colors Blue, Green, Orange, Red and Yellow.



- Wrapped Candy:** It is created by matching 5 candies of same color in L or T shape. Wrapped candy can be made with 5 different colors Blue, Green, Orange, Red and Yellow.



- Color Bomb:** It is created by matching 5 candies of same color in row or column.



5. **Plain Candy + Striped Candy:** When 3 similar color candies lie in same column, or row of grid, they will be crushed. If any of them is a striped candy then the corresponding rows or columns, which contains the striped candies, will be cleared. The decision to clear the row or column will be made according to direction of stripes on candy.
6. **Plain Candy + Wrapped candy:** When 3 similar color candies lie in same column, or row of grid, they will be crushed. If any of them is a wrapped candy then the candies in (3 x 3) square area of board around wrapped candy will also be crushed.
7. **Color Bomb + Plain Candy:** If player swaps color bomb with any plain candy then all candies, of that particular color, will be crushed.
8. **Color Bomb + Color Bomb:** If player swaps color bomb with another color bomb, then all candies from board will be crushed and board will be reloaded with all new candies.



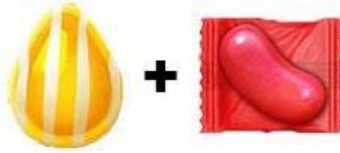
9. **Striped Candy + Color Bomb:** If player swaps color bomb with striped candy, then all candies will be transformed into striped candies of same color as of striped candy. All of them then clear their respective rows/columns.



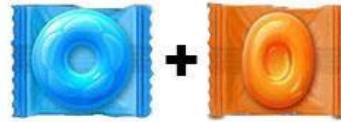
10. **Wrapped Candy + Color Bomb:** If player swaps color bomb with wrapped candy, then all candies will be transformed into wrapped candies of same color of wrapped candy. All candies in (3 x 3) square area, around swapped wrapped candy, will also be crushed.



11. **Striped Candy + Wrapped Candy:** If player swaps striped candy with wrapped candy, then 3 rows and 3 columns centered on the candy that was swapped will be cleared.



- 12. Wrapped Candy + Wrapped Candy:** If player swaps wrapped candy with another wrapped candy, then all candies will be crushed in (5 x 5) square area around swapped candies.



- 13. Winner/Loser:** The winner of this game is decided on the basis of points gained in 20 moves. If gained points are below targeted goal points then player will lose the game.
- 14. User Interaction:** Design an interactive graphical user interface for game. Whenever a new game is started, number and placement of candies should be random. Your game should be user interactive and should provide proper options to player. The information of target point, gained points and remaining moves should be updated accordingly.
- 15. Save\Load Game:** A player can stop game in middle and can continue next time with that game. Therefore, you have to store the state of Game in a file, and next time user will be provided with two options, load new game or play the previous one.

Bonus:

- You can add more frills in the game if you want.

Note:

You can come up with another idea if you want. However, you will need to discuss the idea with me via email or Google meet.

You cannot select the following games:

1. Tic-tac-toe
2. Snakes-and-ladders
3. Checkers

Best Of Luck!